FIREVISION AI

Aam istemaal karne wale ke liye qadam ba qadam: install, roz chalana, login, dashboard ka har hissa, aur har feature ka test, ke screen par kya dikhna chahiye. Test ki boxes yahin tick karein: tick is browser mein yaad rehti hai.

2 windows
Server (API + Dashboard) · AI service

9 tests
Aag demo se camera calibration tak

Safe Test Mode
Test mein asli SMS/Email band

Checklist
Tick karein, progress upar dikhe

Tests: 0 / 31 ho gaye

Ticks saaf karein

A · SHURU KARNE SE PEHLE

1. Ye Guide Kis Ke Liye Hai?

Ye guide **aam istemaal karne wale** ke liye hai, programmer hona zaroori nahi. Is mein qadam ba qadam likha hai:

1. FireVision AI ko pehli dafa computer par kaise **install** karein.
2. Roz kaise **chalayein** (start) aur band karein.
3. Website (dashboard) ka **har hissa** kya karta hai.
4. Har feature ko **khud test** kaise karein, aur test ke baad screen par **kya dikhna chahiye**.
5. Masla aaye to **kya karein**.

FireVision AI ek line mein

FireVision AI building ka **"smart chowkidar"** hai. Camera mein AI aag, dhuwan aur insaan pehchaanta hai. Sensors garmi aur dhuwan napte hain. Khatra ho to screen par alarm aata hai,

SMS/Email/Push jaate hain, sprinklers chal sakte hain, aur naqshe par **mehfooz raasta** dikhta hai.

Guide kaise parhein

- `Is tarah ke dabbe` mein jo likha hai, wo **computer mein type** karna hai (ya copy-paste).
- ✅ = test pass hone par yehi dikhna chahiye.
- ⚠️ = dhyan dein, zaroori baat.
- `<project>` ka matlab hai project ka folder, is computer par: `C:\Users\m_fahad\Desktop\Project`.

> ℹ️ **Note (2026-10-26 se):** project ab **Django** par chalta hai (folder `sentinel-django\`): ek hi server dashboard, API aur live updates deta hai, address **http://localhost:5291**. Purana ASP.NET Core + Angular wala hissa hata diya gaya hai. Zyada tar tasveerein (2026-10-28 se) naye **Django dashboard** ki hain, jo test database, asli AI models aur test video ke saath li gayin. Yeh kuch abhi purane Angular dashboard ki hain: `evacuation-before.png`, `evacuation-camera-people.png`, `camera-safe-danger-zones.png` (aur `swagger.png`, jo ab maujood hi nahi).

3 hifazati usool (pehle parh lein)

> ⚠️ **1.** Ye system **test aur demo** ke liye hai. Ise asli fire alarm ki jagah **na** lagayein. Asli building mein smoke detectors aur fire panel zaroori hain.
>
> ⚠️ **2.** Test se pehle **"Safe Test Mode"** (Chapter 4) zaroor karein, warna test ke dauran **asli SMS aur Email** chale jayenge.
>
> ⚠️ **3.** Kisi **asli sprinkler, alarm ya camera** ko bina ijazat aur bina safe tareeqe ke system se na jorein.

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2. Pehli Dafa: Zaroori Software Install Karna

Ye kaam **sirf ek dafa** karna hai. Har software ki official website se download karein.

#	Software	Kahan se	Kyun
1	**Python 3.13**	python.org (install mein "py launcher" tick rehne dein)	Server (dashboard + API) aur AI service, dono ke liye
2	**SQL Server** 2019 ya naya (Developer, Express ya LocalDB, free)	microsoft.com/sql-server	Database
3	**ODBC Driver 18 for SQL Server**	learn.microsoft.com (search: "ODBC Driver 18 for SQL Server")	Python ko SQL Server se jorne ke liye

.NET SDK, Node.js, Angular CLI ya dotnet-ef ki **ab zaroorat nahi**. (Sirf browser wale automatic tests ke liye Node/npm chahiye, Chapter 17; ye zaroori nahi.)

Check karein ke Python install ho gaya:

```
py -3.13 --version      # Python 3.13.x
```

✅ Version number dikhe, error nahi. ODBC Driver check: Start → "ODBC Data Sources (64-bit)" → **Drivers** tab mein "ODBC Driver 18 for SQL Server" ho.

PowerShell kaise kholein: Start menu → "PowerShell" likhein → **Windows PowerShell** par click.

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3. Pehli Dafa: Project Ka Setup

Ye bhi **sirf ek dafa**.

Qadam 1: `.env` file banana (settings aur passwords)

1. Project folder kholein: `C:\Users\m_fahad\Desktop\Project`.
2. `.env.example` file ki copy banayein aur naam `.env` rakhein (is computer par `.env` pehle se bani hui hai).
3. `.env` ko Notepad mein kholein aur ye bharein:

Line	Kya likhna hai
`Jwt__SigningKey=`	Lambi random chaabi (neeche command se banayein)
`Device__ApiKey=`	Ek aur random chaabi (machines ka password)
`Seed__AdminEmail=`	Login email, jaise `admin@sentinel.local`
`Seed__AdminPassword=`	Apna login password. Khali chhoda to Server pehli dafa ek random password **Server window mein print** karega (sirf ek dafa)

Random chaabi banane ki command (PowerShell):

```
[Convert]::ToBase64String((1..48 | % {Get-Random -Max 256}))
```

Jo text aaye, copy karke `=` ke baad paste karein. `Device__ApiKey` ke liye dobara chalayein.

> ⚠️ `.env` mein passwords hain. Ise **kisi ko na bhejein**, na internet par daalein.

Qadam 2: Server (Django) ke packages

```
cd C:\Users\m_fahad\Desktop\Project\sentinel-django
py -3.13 -m venv .venv
.\.venv\Scripts\python -m pip install -r requirements.txt
```

✅ Kuch minute baad bina laal "ERROR" ke khatam ho, aur `sentinel-django\.venv\` folder ban jaye.

Qadam 3: Database

- `SentinelDb` **pehle se hai** (is computer par hai): kuch nahi karna, server usi ko waise hi istemaal karta hai.
- **Bilkul naya, khali database:** pehle SQL Server mein `SentinelDb` naam ka **khali** database banayein (misaal SQL Server Management Studio → Databases → New Database), phir:

```
cd C:\Users\m_fahad\Desktop\Project\sentinel-django
.\.venv\Scripts\python manage.py create_schema
```

✅ Ye line aaye: `Created the Sentinel tables in 'SentinelDb' …`. Ye saari tables bana deta hai. Agar tables pehle se hon to ye **mana kar deta hai** (data mehfooz rehta hai).

> ⚠️ `SentinelDb` par `manage.py migrate` **kabhi na** chalayein. (System khud bhi ise rok deta hai.)

Doosre PC ka data chahiye (misaal office wala data ghar par)? To `create_schema` ke bajaye `.\database\load.ps1` chalayein (Chapter 5, "Database doosre PC se").

Qadam 4: AI service ka setup

```
cd C:\Users\m_fahad\Desktop\Project\ai-service
py -3.13 -m venv .venv
.\.venv\Scripts\python -m pip install -r requirements.txt
```

- Dono AI models repo ke saath aate hain, kuch download nahi karna.
- Agar `ai-service\models\` mein koi `.onnx` file na ho, to `.\.venv\Scripts\python tools\download_models.py` person model official website se laata hai (sha256 check ke saath).

✅ `ai-service\models\` folder mein ye 3 files hon: `models.json`, `yolox_tiny.onnx`, `fire_smoke_yolox.onnx`.

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4. Safe Test Mode (Test Se Pehle Zaroor)

Masla: is computer ki `.env` mein **asli** Twilio (SMS) aur Email ki settings hain. Test mein aag ya emergency bani to system **asli logon ko asli SMS/Email** bhej dega.

Hal: test ke dauran SMS aur Email band karna

1. `.env` Notepad mein kholein.
2. In lines ke shuru mein `#` laga dein (`#` wali line system parhta hi nahi):

```
#Notifications__Sms__TwilioAccountSid= ...
#Notifications__Sms__TwilioAuthToken= ...
#Notifications__Sms__FromNumber= ...
```

```
#Notifications__Sms__Recipients= ...
#Notifications__Email__SmtpHost= ...
```

3. Save karein. Server chal raha ho to band karke dobara chalayein (Chapter 5).

Is ke baad kya hoga:

- SMS: Live Event Log mein "**skipped (not configured)**". Koi SMS nahi jayega.
- Email: asal mein nahi jayega. Us ka poora text **Server wali PowerShell window** mein print hoga ("console mode"). Test ke liye ye behtareen hai.
- Browser Push: sirf usi browser par jahan aap ne khud **Enable Alerts** dabaya ho.

Test khatam hone ke baad `#` hata dein taake asli alerts dobara chalein.

B · ROZ KA ISTEMAAL

5. Roz Ka Kaam: System Chalana

Sab se aasan: ek click

```
cd C:\Users\m_fahad\Desktop\Project
.\start.ps1
```

Do windows khud khul jati hain (Server = API + Dashboard, aur AI service). ~30 second baad browser mein **http://localhost:5291** kholein. Aag ka demo bhi chahiye to `.\start.ps1 -Simulator` (teesri window simulator ki). Purana `-Django` option bhi chalta hai, lekin ab zaroori nahi.

> Script block ho to: `powershell -ExecutionPolicy Bypass -File .\start.ps1` . Band karne ke liye har window band kar dein.

Ya haath se (har hissa alag window mein)

Windows band na karein jab tak system istemaal ho raha ho.

Window 1: Server (API + Dashboard)

```
cd C:\Users\m_fahad\Desktop\Project\sentinel-django
.\.venv\Scripts\python -m uvicorn config.asgi:application --host 127.0.0.1 --port 5
```

✅ Ye line aaye: `Uvicorn running on http://127.0.0.1:5291`

⚠️ `manage.py runserver` istemaal na karein: wo live updates nahi bhej sakta. Server sirf **ek** hi chalayein.

Window 2: AI Service

```
cd C:\Users\m_fahad\Desktop\Project\ai-service
$env:SENTINEL_DEVICE_KEY = '<yahan Device__ApiKey>'
.\.venv\Scripts\python -m sentinel_ai
```

✅ Ye line aaye: `models loaded: ['person', 'fire_smoke'], missing: []`

Window 3 (sirf demo ke waqt): Simulator

Simulator **nakli** cameras, sensors aur aag banata hai, taake asli aag ke baghair demo ho sake (Chapter 8).

System band karna

Har window mein **Ctrl + C** dabayein, ya window band kar dein. Data database mein mehfooz rehta hai.

Database doosre PC se (office ↔ ghar)

Dono PCs par ek jaisa data rakhne ke liye database ek backup file (`database\SentinelDb.bak`) ban kar git se jata hai.

Kaam khatam karte waqt (jis PC par kaam kiya):

```
cd C:\Users\m_fahad\Desktop\Project
.\database\save.ps1
git add database\SentinelDb.bak
```

```
git commit -m "db: update"
git push
```

Doosre PC par kaam shuru karte waqt (pehle Server band karein):

```
git pull
.\database\load.ps1
```

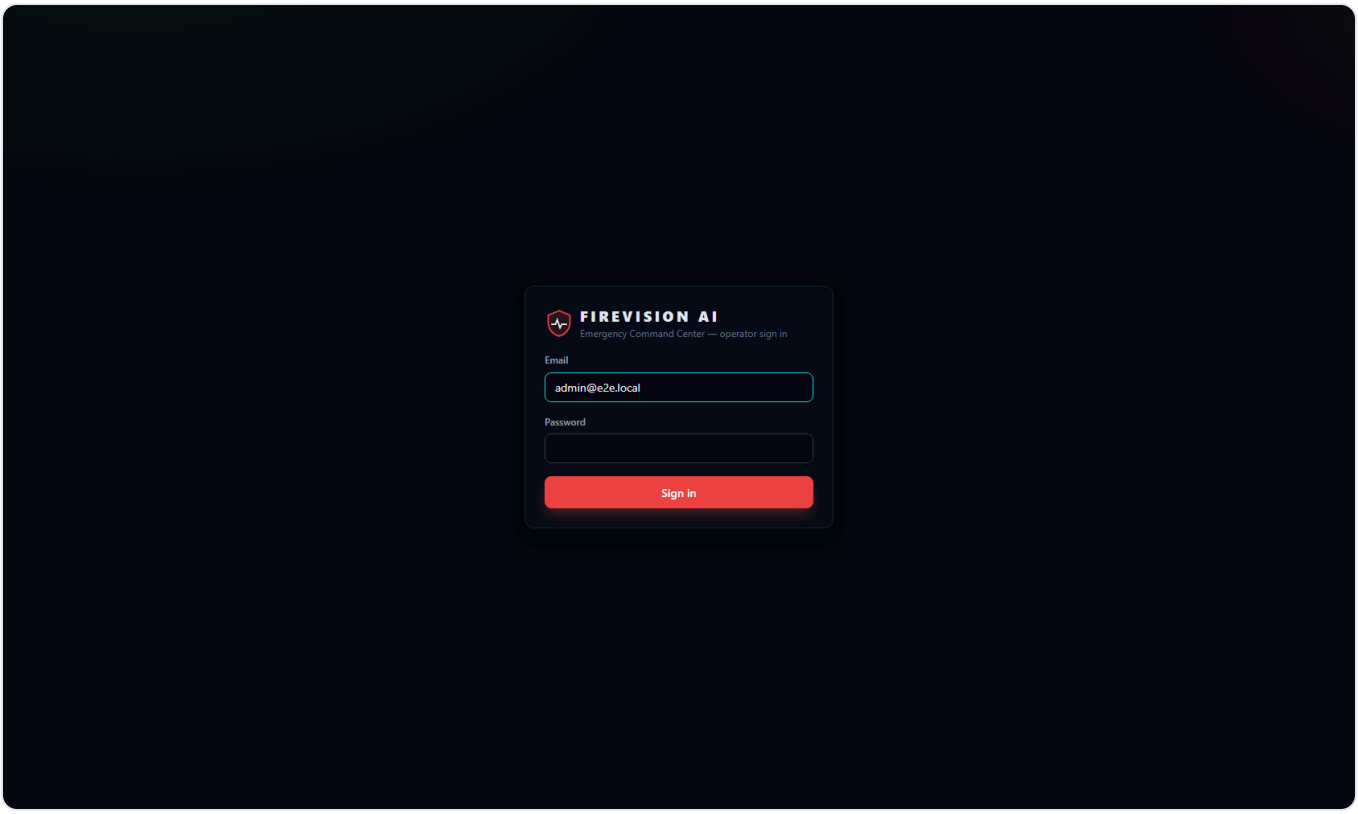
"yes" likhein. ✅ "Restored SentinelDb..." aaye. Database na ho to ban jata hai, ho to **poora badal** jata hai.

> ⚠️ Ek waqt mein **ek hi PC** par kaam karein. Dono jagah data badla aur save/load bhool gaye, to ek jagah ka kaam mit jayega. Login wohi email/password chalta hai jo pehle PC par tha.

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6. Login Karna

1. Browser (Chrome/Edge) kholein: **http://localhost:5291**
2. **Email:** `.env` wala `Seed__AdminEmail` (jaise `admin@sentinel.local`).
3. **Password:** `.env` wala `Seed__AdminPassword`. Khali chhoda tha to pehli dafa Server ki window mein **sirf ek dafa** print hua tha ("GENERATED password").
4. **Sign in** dabayein.

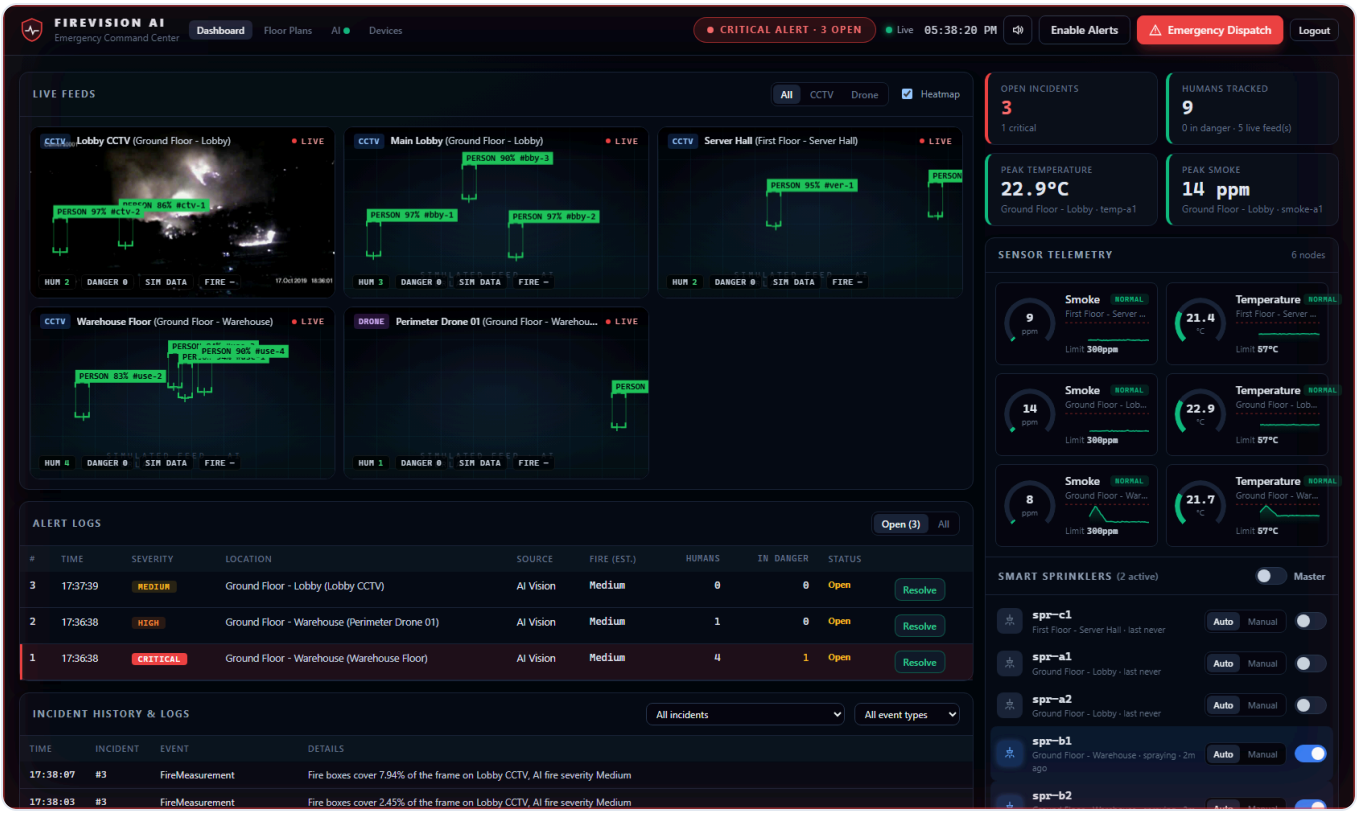


✅ Dashboard khul jaye aur upar dayen taraf **Live** (hara nukta) likha ho.

Ghalat password: laal message aayega. Login 8 ghante tak chalta hai, phir dobara login.

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7. Dashboard Ka Tour (Har Hissa Kya Hai)



Upar ki patti (Top bar)

Cheez	Matlab
Dashboard · Floor Plans · AI	Teen pages ke beech jana
Beech ka pill	Hara **"ALL SYSTEMS NOMINAL"** = sab theek **aur** AI chal rahi hai. Peela **"MONITORING DEGRADED · AI OFFLINE"** (ya AI MODEL MISSING / CAMERA OFFLINE / AI DATA STALE / LIVE LINK DOWN; har 10 second, aur AI ke frames 8 second ruk jayein to foran). Incident khula ho to wajah saath likhi jati hai, jaise "ELEVATED RISK · 1 OPEN · AI OFFLINE" = koi incident nahi, lekin **nigrani adhoori hai**: is ka matlab "aag nahi" nahi, balke "dekha hi nahi ja raha". Peela "ELEVATED RISK" / laal "CRITICAL ALERT" = khule incidents
Live / Offline	Server se live connection hai ya nahi
Ghari	Abhi ka asli waqt, AM/PM ke saath (Windows jaisa)
button	Alarm ki awaaz band/chalu. Koi bhi incident khula ho to alarm **lagatar** bajta hai, jab tak incident **Resolve** na karein ya ye button na dabayein
Enable Alerts	Is browser par Push notifications on karna

Cheez	Matlab
Emergency Dispatch (laal)	Haath se emergency elaan (Chapter 12)
Logout	Bahar nikalna

Dashboard ke hisse

Hissa	Kya dikhata hai	Aap kya kar sakte hain
KPI patti (4 dabbe)	Khule incidents, insaan, sab se zyada garmi, sab se zyada dhuwan	Sirf dekhna
Live Feeds (cameras)	Har camera, AI ke boxes aur zones	All/CCTV/Drone filter, Heatmap on/off, camera par click = bara
Sensor Telemetry	6 sensors ke gauges (dhuwan ppm, garmi °C)	Dekhna
Smart Sprinklers	Har sprinkler ON/OFF, Auto/Manual	Switch, Auto/Manual, Master
Alert Logs	Incidents ki list	Open/All filter, **Resolve**
Live Event Log	Har waqia (SMS gaya, sprinkler chala...)	Dekhna
Incident History & Logs	Aag ka graph aur history	Incident aur event type ka filter
Evacuation Route (estimate)	Naqsha, mehfooz raasta, aur aag lagne par **risk heatmap**	Naqshe par click = raasta

Camera tile par rang

Rang	Matlab
Laal dabba (dhadakta)	Aag
Gray dabba	Dhuwan
Hara dabba	Mehfooz insaan

Rang	Matlab
🟠 Amber "IN DANGER"	Insaan khatre ke zone mein
🔴 Laal dashed shakal	Danger zone
🟢 Hari shakal	Safe zone (assembly point)
SIM DATA chip	Data nakli (test) hai
NO AI DATA	AI se 1.5 second se data nahi aaya

Severity (khatre ka darja)

Low (halka) → **Medium** → **High** (zyada) → **Critical** (sab se khatarnak).

C · HAR FEATURE KA TEST

8. Test 1: Aag Ka Demo (Simulator)

Maqsad: dekhna ke aag lagne par poora system kaise react karta hai.

> ⚠️ Pehle **Safe Test Mode** (Chapter 4) karein. Is demo mein insaan danger zone mein aate hain, jo **emergency** hai, aur SMS/Email chalte hain.

Qadam:

1. Window 1 (Server) chal rahi ho. Login karein.
2. `.env` se apni `Device__ApiKey` copy karein.
3. Nayi PowerShell window (Window 3):

```
cd C:\Users\m_fahad\Desktop\Project\sentinel-django
.\.venv\Scripts\python tools\simulator.py --device-key <yahan Device__ApiKey> --ign
```

(Ya seedha `.\start.ps1 -Simulator`: ye simulator ki window khud khol deta hai.)

4. Browser mein Dashboard dekhein.

✅ **Kya dikhna chahiye (1–2 minute mein):**

- [ ] Cameras par boxes hilte hue. **SIM DATA** chip.
- [ ] **Warehouse Floor** (Ground Floor - Warehouse) camera par laal aag ka dabba aur laal danger zone, hari "assembly point".
- [ ] Upar pill laal: **CRITICAL ALERT**. Alarm **lagatar** bajta hai (Resolve karne ya 🔊 dabane tak).
- [ ] **Alert Logs** mein naya incident, Location "Ground Floor - Warehouse", Severity High/Critical.
- [ ] **Live Event Log** mein lines: incident khula, SMS "skipped", email (Server window mein print).
- [ ] Ground Floor - Warehouse ke garmi/dhuwan gauges barhte hue.
- [ ] Ground Floor - Warehouse ke **Auto** sprinklers **khud chal jaate hain** ("spraying"). Demo setting (`SENTINEL_DEBUG` , ya `Alerts__AutoSprinklerOnSimulatedData=true`) mein simulated data par bhi chalte hain; asli hardware nahi, sirf database.

Band karna: Window 3 mein **Ctrl + C**.

Camera lost check (simulator band karne ke baad):

- [ ] ~10–12 second baad **Alert Logs** mein har camera ke liye ek incident, Source **Camera lost**: "Camera lost: no frames from … for 10 s".
- [ ] Ground Floor - Warehouse ke cameras (jahan aag thi) **High**, baqi **Medium**.
- [ ] Simulator dobara chalayein: Live Event Log mein "Camera … is sending frames again". Phir purane "Camera lost" incidents **Resolve** kar dein.

Simulator ke options:

Option	Kaam	Default
`--ignite "Ground Floor – Lobby"`	Us zone mein foran aag (kai dafa likh sakte hain)	—
`--fps 2`	Har camera ke frames per second	5
`--fire-probability 0`	Khud se random aag band	0.004
`--telemetry-interval 5`	Sensors kitne second baad	2

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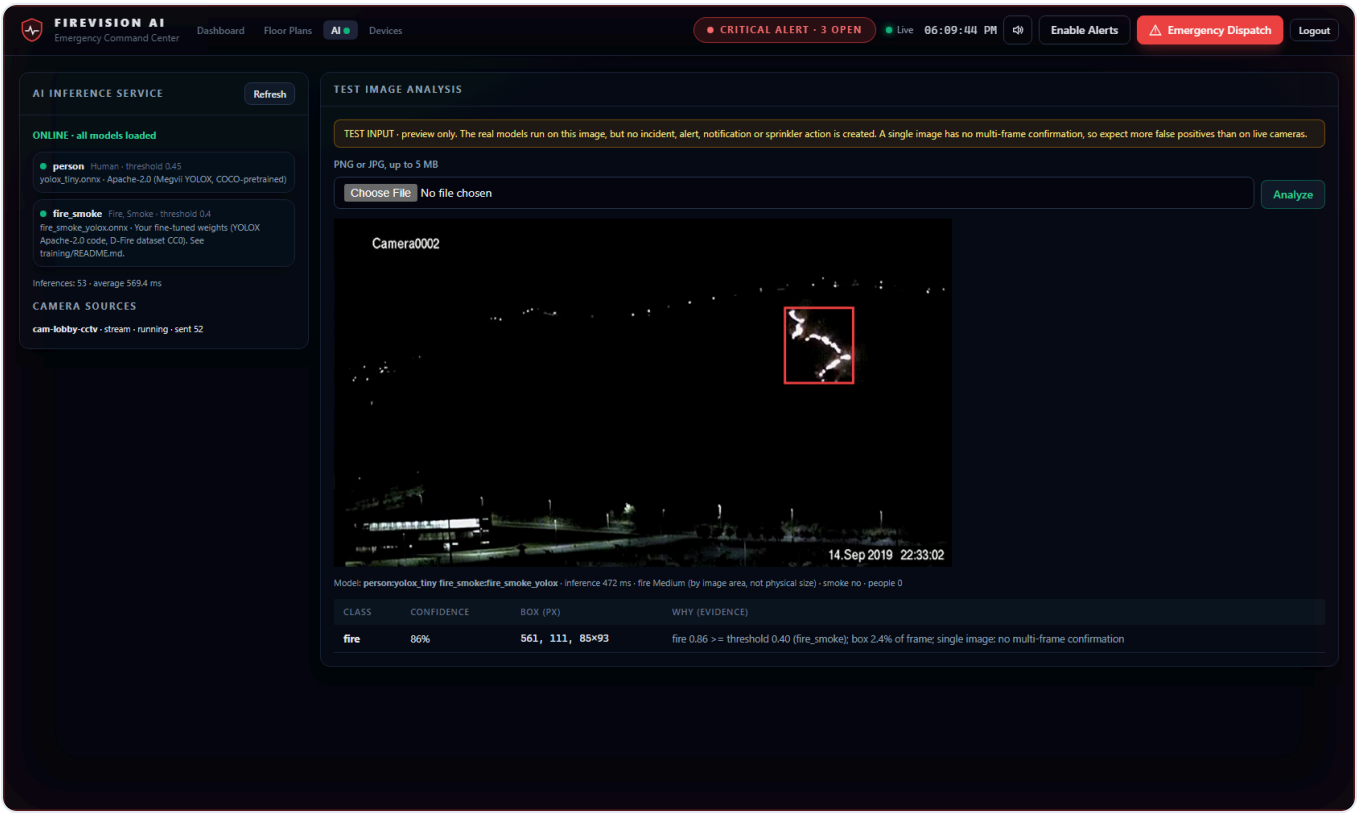
9. Test 2: AI Ko Photo Dikhana (AI Page)

Maqsad: dekhna ke AI models asal mein aag, dhuwan aur insaan pehchaante hain.

Test photo kahan se: aag wali koi bhi photo (JPG/PNG, 5 MB tak). D-Fire dataset ki test photos `Desktop\sentinel-colab\D-Fire.zip` ke andar `test\images\` mein hain (misaal `AoF07831.jpg`). Insaan wali koi bhi photo.

Qadam:

1. Window 1 (Server) aur Window 2 (AI service) chal rahi hon.
2. Upar menu mein **AI** par click.
3. Baen taraf dekhein: hara **ONLINE · all models loaded**, dono models (`person` , `fire_smoke`) ke aage hara nukta.
4. Daen taraf **Choose image…** dabayein aur photo chunein.
5. 1–3 second intezar.



✅ Kya dikhna chahiye:

- [ ] Photo par dabbe: aag **laal**, dhuwan **gray**, insaan **hara**.
- [ ] Neeche line: model ke naam, kitne ms lage, "fire Medium", "smoke yes", "people 0".

- [ ] Table: Class, Confidence (jaise 83%), Box, aur **Why (evidence)** yaani AI ne kyun kaha.
- [ ] Bina aag wali photo par aag ka dabba **na** aaye.

Yaad rakhein:

- Ye sirf **preview** hai: koi incident, alert ya sprinkler **nahi** chalta.
- AI se kabhi galti ho sakti hai. Kuch na mile to iska matlab ye nahi ke jagah mehfooz hai.

Masla:

Screen par	Matlab	Hal
Laal **OFFLINE**	AI service band hai	Window 2 chalayein
Peela "some models missing"	Koi model file nahi	`ai-service\models\` check (Chapter 3, Qadam 4)
"sha256 mismatch"	Model file badli ya kharab	Sahi file dobara copy karein

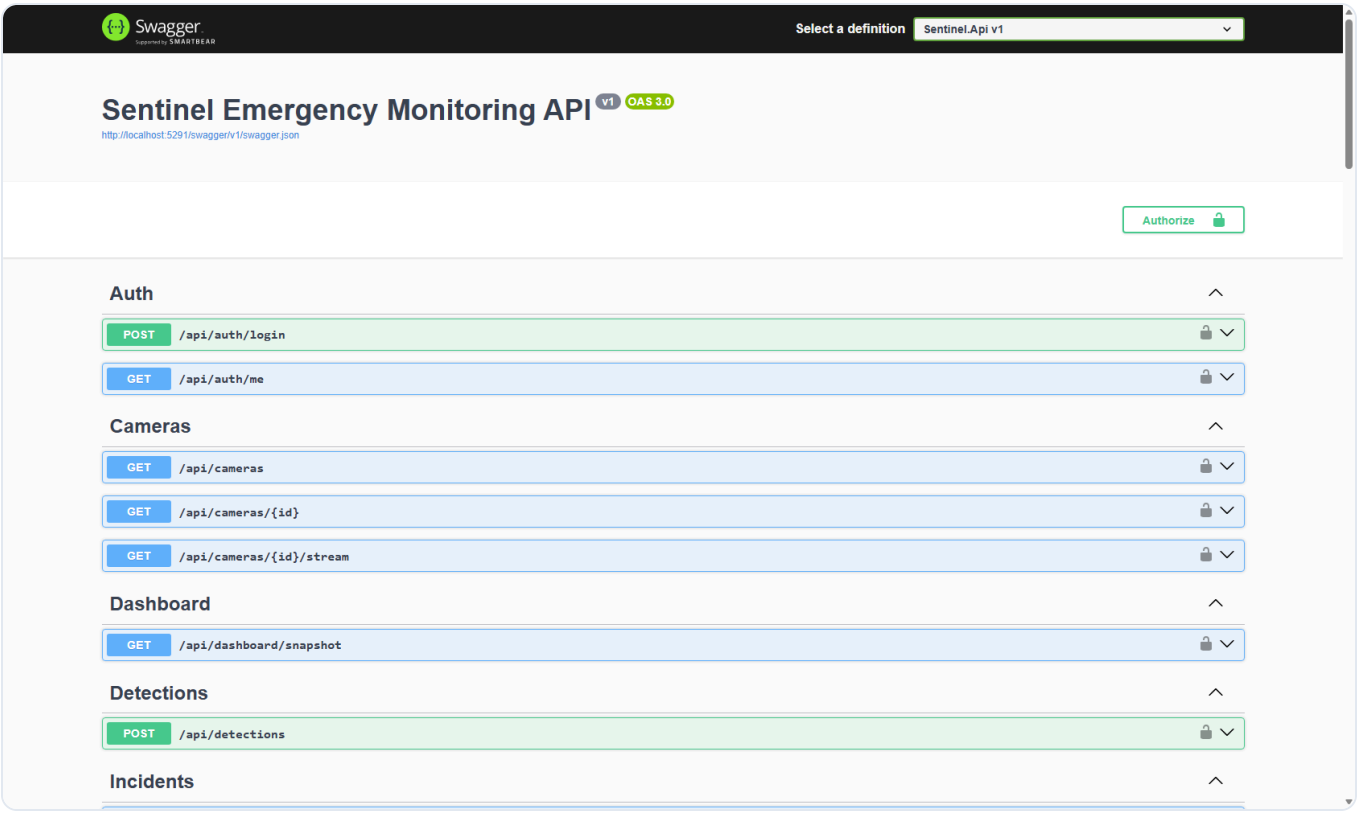
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10. Test 3: Sensor Ko Garam Karna (PowerShell)

Maqsad: dekhna ke garmi barhne par Warning, Alarm aur Critical sahi waqt par aate hain.

Pehle ye kaam Swagger page se hota tha; Django mein **Swagger page nahi hai**. Ab hum PowerShell se seedha sensor ki reading bhejte hain, bilkul waise jaise asli sensor (machine) bhejta hai: `X-Device-Key` header mein `.env` wali `Device__ApiKey`.

Neeche purani tasveer sirf hawale ke liye hai (Swagger ab nahi hai):



Qadam A: tayyari

1. Window 1 (Server) chal rahi ho aur browser mein Dashboard khula ho.
2. Nayi PowerShell window kholein aur apni device key rakhein:

```
$key = '<yahan Device__ApiKey>'
```

Qadam B: garmi bhejna

1. Sensor `temp-b1` (Ground Floor - Warehouse ka temperature sensor; threshold 57°C). Ye command ek reading bhejti hai (yahan `30`):

```
Invoke-RestMethod -Method Post -Uri http://localhost:5291/api/iot/sensors/temp-b1/r
```

✅ Jawab mein sensor ka haal aaye: `id` = `temp-b1`, `currentValue` = `30`, `state` = `Normal`. Ghalat key par **401** error.

2. Ab `-Body` mein ek ek karke ye values bhejein (sirf number badlein), har dafa Dashboard dekhein:

Value	Kya hona chahiye	Hisaab
`{ "value": 30 }`	✅ Normal, kuch nahi	45.6 se kam
`{ "value": 50 }`	✅ Gauge **peela** (Warning), Live Event Log mein "Normal → Warning", incident nahi	≥ 57 × 0.8 = 45.6
`{ "value": 60 }`	✅ **High** incident + Ground Floor - Warehouse ke **Auto** sprinklers ON	≥ 57
`{ "value": 75 }`	✅ **Critical** incident + **emergency** notifications + sprinklers	≥ 57 × 1.25 = 71.25

Sensor IDs (ye demo data ke sensors hain, jo pehli dafa start par bante hain jab demo data on ho, `Seed__DemoData` ya `SENTINEL_DEBUG`; na milein to jawab mein "Sensor '...' not found"):

Zone	Dhuwan (threshold 300 ppm)	Garmi (threshold 57°C)
Ground Floor - Lobby	`smoke-a1`	`temp-a1`
Ground Floor - Warehouse	`smoke-b1`	`temp-b1`
First Floor - Server Hall	`smoke-c1`	`temp-c1`

Dhuwan ke liye Critical = 300 × 1.5 = **450 ppm**.

> ⚠️ Sensor ka data **asli** maana jata hai (SIMULATED nahi), is liye Alarm/Critical par Auto sprinklers **chal jate hain**. Is project mein sprinklers sirf database mein hain, koi asli hardware nahi. Test ke baad unhein OFF karein (Chapter 11).

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11. Test 4: Sprinklers

Dashboard → **Smart Sprinklers** panel.

Kaam	Kaise	✅ Kya dikhe
Ek sprinkler ON/OFF	Us ka switch dabayein	Switch badle, Live Event Log mein line
Auto mode	"Auto" button	Alarm/emergency par system khud chala sakta hai
Manual mode	"Manual" button	System khud nahi chalayega, sirf aap
Master	Upar "Master" switch → confirm	Saare sprinklers ek saath ON/OFF

Check: ek sprinkler **Manual** karein, phir Test 3 mein `75` bhejein. ✅ Manual wala sprinkler **OFF** rehna chahiye. Auto wale ON.

Yaad rakhein: sprinkler khud OFF **nahi** hota. Aag ke baad aap OFF karein.

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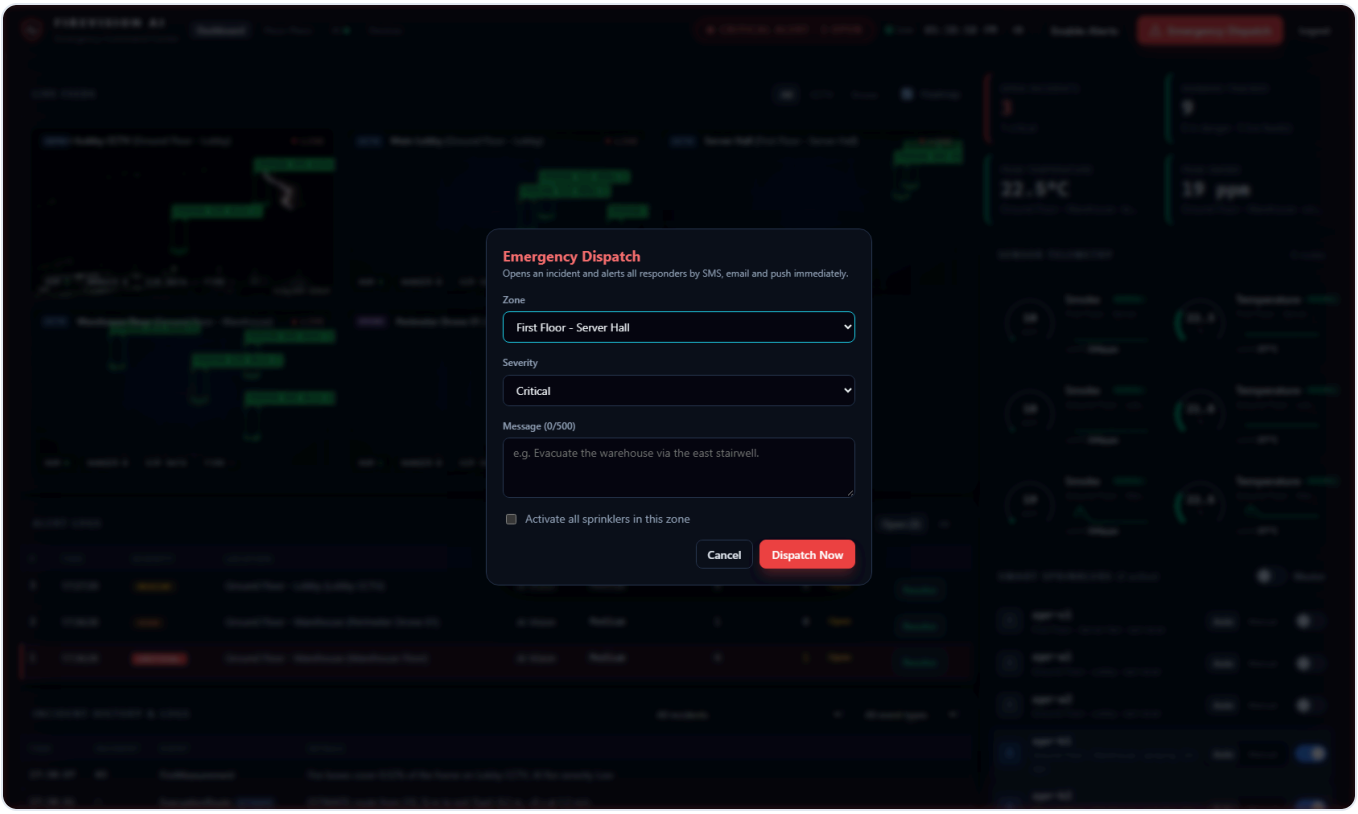
12. Test 5: Emergency Dispatch Aur Resolve

Maqsad: haath se emergency elaan karna, phir incident band karna.

> ⚠️ Dispatch **foran** SMS/Email/Push bhejta hai. Pehle Safe Test Mode (Chapter 4).

Qadam:

1. Upar laal **Emergency Dispatch**.
2. Form:
   - **Zone:** jaise `Ground Floor – Warehouse`
   - **Severity:** Low / Medium / High / Critical
   - **Message:** jaise "Test: Ground Floor - Warehouse khaali karein" (500 huroof tak)
   - **Activate all sprinklers in this zone:** tick sirf agar sprinklers chalane hon
3. **Dispatch Now** dabayein.



✅ **Kya dikhna chahiye:**

- [ ] Alert Logs mein naya incident (Source: Manual).
- [ ] Live Event Log: SMS skipped, email Server window mein.

Resolve (band karna):

1. **Alert Logs** mein us incident ki line par **Resolve**.
2. ✅ Incident "Open" list se hat jaye ("All" mein Resolved dikhe). History mein `Resolved` log.

---

13. Test 6: Naqsha Banana Aur Evacuation Raasta

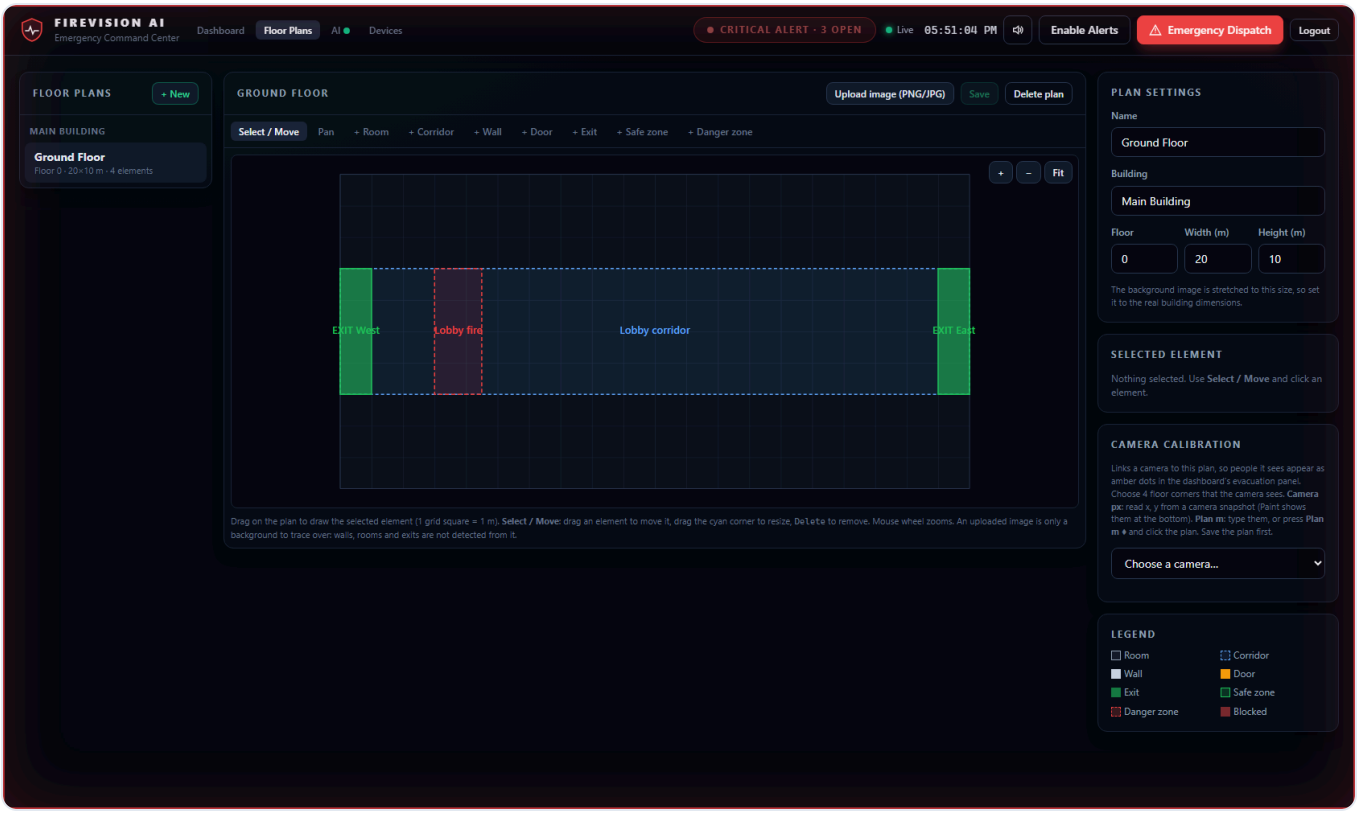
A. Naqsha banana

1. Upar **Floor Plans** → **+ New**.
2. Daen taraf **Plan settings**: Name ("Ground Floor"), Building ("Warehouse"), Floor (0), **Width aur Height meters mein** (jaise 30 × 20).
3. (Ikhtiyari) **Upload image (PNG/JPG)**: apne naqshe ki photo background mein (5 MB tak).

4. Toolbar se drag karke banayein:

Tool	Kya banana hai
+ Room	Kamre / halls
+ Corridor	Gallery
+ Wall	Deewarein (kamron ke beech)
+ Door	Deewar par darwaza (darwaze ke baghair raasta deewar paar nahi karta)
+ Exit	Bahar nikalne ke darwaze (kam az kam ek)
+ Safe zone	Assembly point
+ Danger zone	Boiler room / khatarnak jagah

5. **Select / Move** se kisi cheez par click → daen panel mein naam, size badlein. Delete key = mitana.
6. Danger zone chunein → **Linked camera/sensor zone** mein list se `Ground Floor - Warehouse` chunein (likhne ki zaroorat nahi, spelling galat nahi hogi) (Ground Floor - Warehouse mein incident khula ho to ye khud active).
7. **Save**. ✅ "Floor plan saved" message, title se "unsaved" hat jaye.



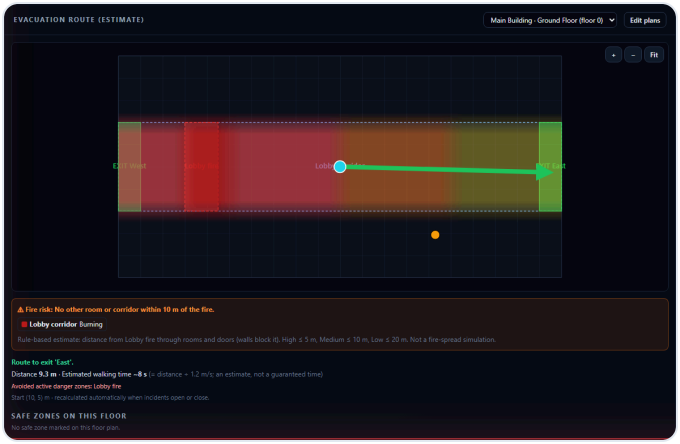
Zoom: mouse wheel ya + / − / Fit. Ek khana = 1 meter.

B. Raasta dekhna

1. **Dashboard** → neeche **Evacuation Route (estimate)** panel → floor plan chunein.
2. Naqshe par wahan **click** karein jahan insaan khara hai.
3. ✅ **Hari teer** qareeb tareen exit tak, "Distance 14.8 m · ~12 s" jaisa likha.

C. Aag mein raasta badalna

1. Ground Floor - Warehouse ke danger zone ke paas wala raasta dekh lein.
2. Test 1 (simulator `--ignite "Ground Floor – Warehouse"`) ya Test 3 (`temp-b1` = 75) chalayein.
3. ✅ Danger zone laal/active, aur raasta **khud badal** kar doosre exit ki taraf, danger se kam az kam 1 m door.
4. Koi raasta na bache to ✅ saaf message: "No safe route..."

Pehle	Aag ke baad
	

Yaad rakhein: waqt sirf andaza hai (faasla ÷ 1.2 m/s). Dhuwan, bheer, seedhiyan shamil nahi.

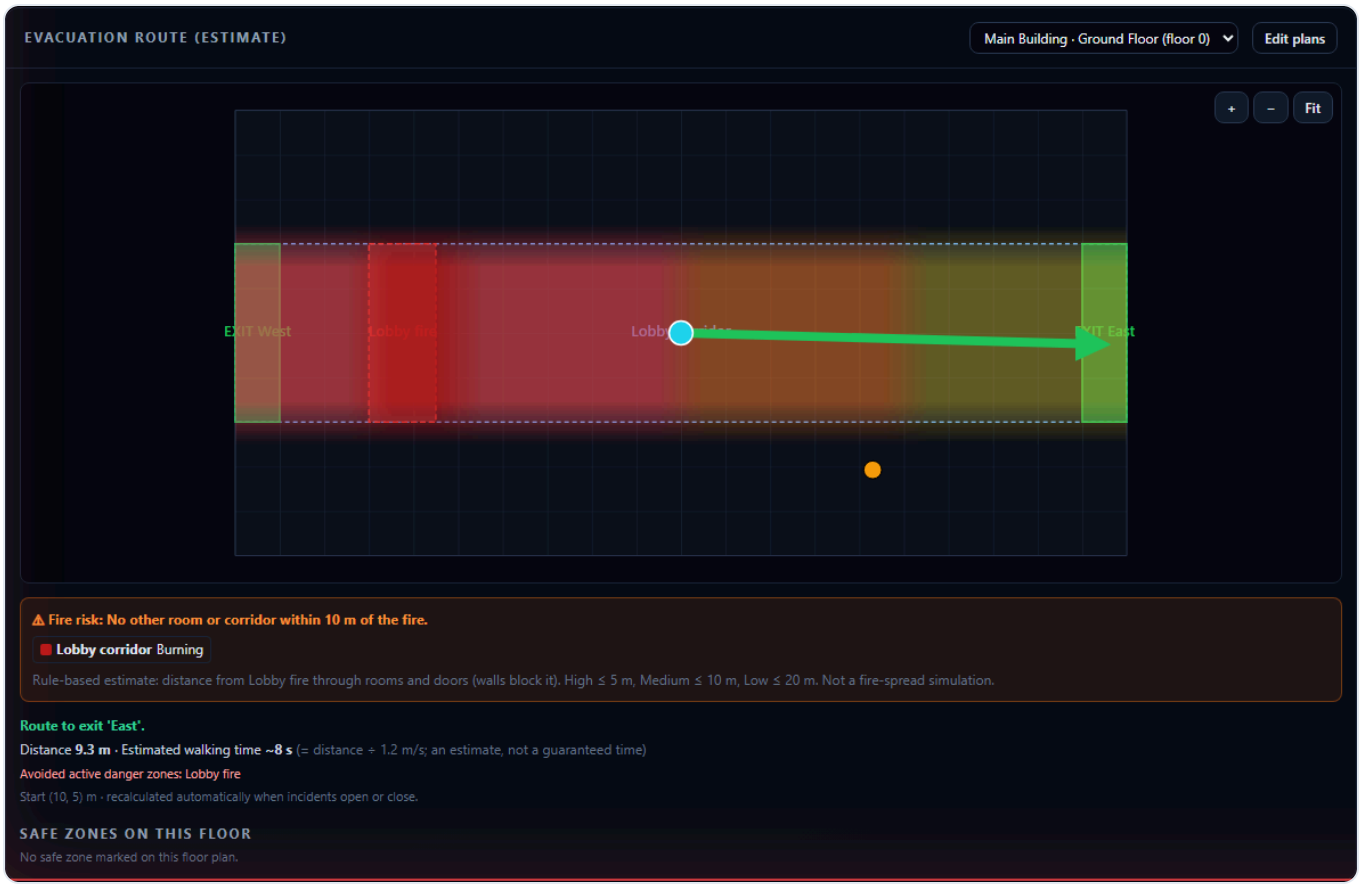
D. Aag kahan phailegi (risk heatmap)

Maqsad: aag lagne par dekhna ki system bataye **agle kaun se kamre khatre mein hain**.

1. Naqshe mein kam az kam 2 kamre, unke beech deewar aur ek **darwaza** hon. Danger zone ka **Linked zone** = `Ground Floor – Warehouse` (A hisse wala naqsha).
2. Ground Floor - Warehouse mein incident kholein: **Emergency Dispatch** (Ground Floor - Warehouse), ya simulator `--ignite "Ground Floor – Warehouse"`.
3. Dashboard → **Evacuation Route** → wohi naqsha.

✅ **Kya dikhna chahiye:**

- [ ] Naqshe par rang: aag wala kamra **gehra laal**, paas wale **laal/narangi**, door wale **peele**.
- [ ] Jis kamre ke aur aag ke beech **poori deewar** hai (darwaza nahi), wo **bina rang** ke (mehfooz).
- [ ] Naqshe ke neeche: "⚠ Fire risk: Next at risk: … (x m from the fire, Medium)" aur har kamre ka darja.
- [ ] Incident **Resolve** karein: heatmap aur list khud gayab.



Yaad rakhein: ye **andaza** hai (darwazon ke raaste faasla). Asal aag hawa, saman aur dhuwen se alag tarah bhi phail sakti hai.

E. Camera ko naqshe se jorna (camera ka insaan khud naqshe par)

Maqsad: camera mein dikha insaan naqshe par khud **amber nuqta** ban kar aaye, aur us par click se raasta nikle.

Ek dafa ki tayyari: camera ki ek **snapshot** (photo) jis mein farsh ke 4 kone saaf dikhein, misaal kamre ke 4 kone.

1. **Floor Plans** → naqsha kholein → pehle **Save** (title par "unsaved" na ho).

2. Daen taraf **Camera calibration** → camera chunein (misaal `Warehouse Floor (cam-warehouse)`).
3. Snapshot ko **Paint** mein kholein. Pehle kone par mouse rakhein: Paint neeche x, y pixel dikhata hai. Ye **Camera px** mein likhein.
4. Usi kone ke liye **Plan m** ⌖ dabayein aur naqshe par wohi kona click karein (ya meters khud likhein).
5. Baqi 3 kono ke liye yehi. Koi 3 kone **ek seedhi line** mein na hon.
6. **Save calibration**. ✅ "Camera calibrated" message, aur list mein camera ke aage "calibrated here".



Check:

1. Simulator chalayein (Test 1) ya AI camera (Test 8).
2. Dashboard → **Evacuation Route** → wohi naqsha.
3. ✅ Naqshe par **amber nuqte**: camera ke insaan, unke pairon ki jagah par.

4. Kisi nuqte par click → ✅ **hari teer** usi jagah se exit tak.



Masle:

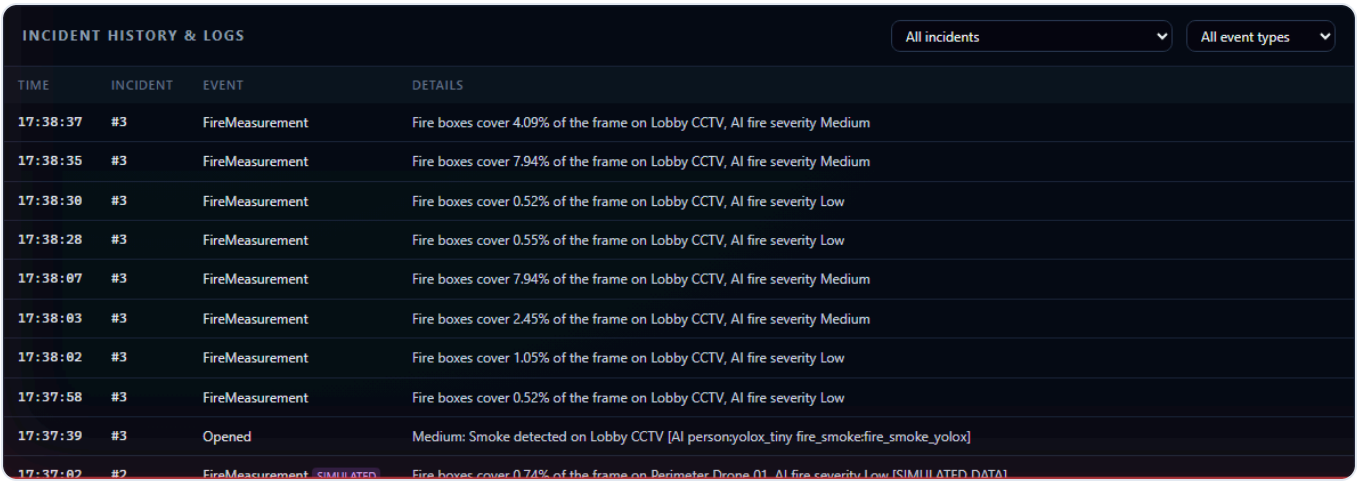
Screen par	Matlab	Hal
"Three of the 4 points lie on one line"	3 kone ek line mein	Alag alag kone chunein
"A point lies outside…"	Naqshe se bahar ka point	Naqshe ke andar ke meters
"Save the plan first"	Naqsha save nahi	Pehle **Save**
Nuqte nahi aate	Camera se frames nahi, ya ghalat naqsha chuna	Simulator/AI chalayein, wohi naqsha chunein
Nuqte ghalat jagah	Points ki tarteeb ulti (camera ka kona 1 ≠ naqshe ka kona 1)	Dono taraf **ek hi kona** ek hi number par

> ⚠️ Camera **hila** to dobara calibrate karein. Nuqta andaza hai, aur seedhiyon par khade insaan ke liye ghalat ho sakta hai.

---

14. Test 7: Incident History Aur Logs

1. Dashboard → **Incident History & Logs**.
2. Upar se ek **incident** chunein (ya "All incidents"), aur event type ("All event types").
3. ✅ Dikhe:
   - ☐ Camera wale incident par **aag ka graph** (frame ka %), sensor wale par **sensor ka graph**.
   - ☐ Logs: `Opened`, `Escalated`, `FireMeasurement`, `PersonInDanger`, `EvacuationRoute`, `Resolved`.
   - ☐ Test data par jamni **SIMULATED**, raaste par neela **ESTIMATE**.



---

15. Test 8: AI Ko Video Se Camera Ki Tarah Chalana

Maqsad: AI ko lagatar frames dena, jaise asli camera, aur dekhna ke wo khud incident banata hai.

Sab se aasan: Devices page se (Admin)

1. Menu → **Devices** → **+ Camera**.
2. Id `cam-gate`, Name "Main Gate", Zone list se chunein, **Stream link** mein camera ka `rtsp://…` (ya `http://…` video), format **Simulated** → **Save**.
3. AI service `.\start.ps1` se chal rahi ho to **1 minute ke andar** ye camera khud uthati hai.

✅ **Kya dikhna chahiye:**

- [ ] Dashboard par "Main Gate (Ground Floor - Gate)" tile.
- [ ] AI page → Camera sources mein `cam-gate · stream · running`.
- [ ] Aag dikhe to Alert Logs mein "Main Gate" ka incident.
- [ ] Camera delete karein to ~1 minute mein AI page se bhi hat jaye.

> Dashboard par link `rtsp://***@…` dikhta hai: password sirf AI service ko jata hai.

Ya `config.json` se (video file ke liye)

Qadam:

1. Aag ki ek **video file** (MP4/AVI) computer par rakhein.
2. `ai-service\config.example.json` ki copy banayein, naam `config.json`.
3. Notepad mein `cameras` hissa aisa karein (apna path likhein, `\` ki jagah `/`):

```
{
  "api": "http://localhost:5291",
  "listen": "127.0.0.1:8008",
  "cameras": [
    { "cameraId": "cam-warehouse", "source": "C:/Users/m_fahad/Videos/fire-test.mp4
  ]
}
```

4. Window 2 (AI service) band karein (Ctrl + C), phir device key de kar dobara chalayein:

```
cd C:\Users\m_fahad\Desktop\Project\ai-service
$env:SENTINEL_DEVICE_KEY = "<yahan Device__ApiKey>"
.\.venv\Scripts\python -m sentinel_ai
```

✅ **Kya dikhna chahiye:**

- [ ] AI page → Camera sources: `cam-warehouse · file · SIMULATED / TEST INPUT · running · sent …`
- [ ] Dashboard ke Warehouse camera par AI ke boxes.
- [ ] Aag **3 frames** tak lagatar dikhe to incident, **SIMULATED DATA** label ke saath.
- [ ] Demo setting mein Auto sprinklers khud chal jaate hain (simulated data par bhi).

Yaad rakhein:

- Video **file** hamesha SIMULATED maani jati hai.
- Agar video mein manzar har frame badalta ho (photos ka slideshow), to aag "confirm" nahi hogi. `fps` barha kar (jaise 4) dobara try karein.
- Is purane computer par kul ~1–2 frame per second ka budget hai.
- **Asli camera** (`rtsp://…`) jorne se pehle Safe Test Mode aur ijazat zaroori. Camera ka password config mein nahi, environment variable mein rakhein.

---

16. Test 9: Browser Notifications

1. Upar **Enable Alerts** dabayein → browser poochhe to **Allow**.
2. Test 1 ya Test 5 chalayein.
3. ✅ Windows ke kone mein FireVision AI ka notification aaye (dashboard ka tab band ho tab bhi, jab tak browser background mein chal raha ho).

Ye sirf tab kaam karta hai jab `.env` mein `Notifications__WebPush__…` (VAPID) keys bhari hon **aur** server ke venv mein `pywebpush` install ho:

```
cd C:\Users\m_fahad\Desktop\Project\sentinel-django
.\.venv\Scripts\python -m pip install pywebpush
```

VAPID keys kisi bhi tool se banayein, misaal (Node.js ho to): `npx web-push generate-vapid-keys`. Keys `.env` mein daal kar Server dobara chalayein.

D · MASLE, HIFAZAT AUR CHECKLIST

17. Automatic Tests (Ek Click Mein Sab Check)

Ye tests code khud check karte hain. Koi test **fail** nahi hona chahiye. Ye apne **alag, waqti** `test_*` databases istemaal karte hain aur `.env` nahi parhte, is liye asli data aur asli SMS/Email mehfooz rehte hain.

```
cd C:\Users\m_fahad\Desktop\Project\sentinel-django
.\.venv\Scripts\python manage.py test --noinput   # ✅ 79 tests, sab pass (2026-10-

# Browser tests (installed Edge). Pehli dafa sirf ek baar (is ke liye Node/npm chahi
cd e2e; npm install --offline; cd ..
.\.venv\Scripts\python e2e\run.py                 # ✅ 10/10 pass (2026-10-28)

# Asli AI service ke saath poora camera lifecycle (~8 minute):
.\.venv\Scripts\python e2e\lifecycle.py           # 2026-10-28: 3 runs mein 15/18, 1

cd C:\Users\m_fahad\Desktop\Project\ai-service
.\.venv\Scripts\python -m unittest discover -s tests    # ✅ Ran 25 tests ... OK
```

AI ki accuracy dobara napna (D-Fire test folder ka path chahiye, ~20 minute):

```
.\.venv\Scripts\python tools\evaluate.py --data <D-Fire\test folder> --classes 0=Sm
```

✅ Threshold 0.4 par fire image recall ~0.93 aur smoke ~0.92 aana chahiye.

---

18. Masle Aur Un Ka Hal

Masla	Wajah	Hal
Top bar par **Offline / Reconnecting**	Server band hai	Window 1 chalayein
Login par "network error"	Server band, ya port 5291 mein masla	Window 1 check, `http://localhost:5291/health` kholein: `"status":"ok"` aana chahiye
Ghalat password	`.env` ka password	`.env` mein `Seed__AdminPassword` dekhein. Users table pehle se ho to wohi purana password

Masla	Wajah	Hal
Server SQL Server se connect nahi hota (connection / login timeout error)	SQL Server band	Start → "Services" → **SQL Server (MSSQLSERVER)** → Start
Server error: "Invalid object name" ya "Invalid column name ..."	Database khali ya purana (tables poori nahi)	Khali database ho to `.\.venv\Scripts\python manage.py create_schema` (Chapter 3). Purana database ho to doosre PC se `.\database\load.ps1`. `manage.py migrate` na chalayein
Server: ODBC / "Data source name not found" / driver error	ODBC Driver 18 install nahi	"ODBC Driver 18 for SQL Server" install karein (Chapter 2)
Cameras par **NO AI DATA**	AI/simulator data nahi bhej raha	Simulator (Test 1) ya AI camera (Test 8) chalayein
AI page **OFFLINE**	AI service band	Window 3 chalayein
AI "some models missing"	`.onnx` file nahi	Chapter 3, Qadam 4
AI camera "API HTTP 401"	Device key ghalat ya nahi di	`$env:SENTINEL_DEVICE_KEY` = `.env` ka `Device__ApiKey`
Simulator "401"	Device key ghalat	`--device-key` mein sahi key
`py -3.13`: "not recognized" ya Python 3.13 nahi mila	Python 3.13 install nahi ya PowerShell purani	Chapter 2, phir **nayi** PowerShell kholein
`start.ps1`: "sentinel-django\.venv is missing" ya `.\.venv\Scripts\python` nahi mila	Server ka venv nahi bana	Chapter 3, Qadam 2 (`py -3.13 -m venv .venv` + `pip install`)
Port 5291 "already in use" / "address already in use"	Server pehle se chal raha hai	Purani Server window band karein, ya computer restart

Masla	Wajah	Hal
SMS/Email nahi aate	Safe Test Mode on, ya settings ghalat	`#` hatayein (Chapter 4), Live Event Log mein error dekhein
`load.ps1`: "Cannot open server" ya "network-related"	SQL Server band, ya Express naam se install	SQL Server start karein; Express ho to `.\database\load.ps1 -Server "localhost\SQLEXPRESS"` aur `.env` mein bhi wohi naam
`load.ps1`: version wala error	Doosre PC par SQL Server purana	SQL Server 2022 ya naya install karein
PowerShell: "running scripts is disabled"	Script chalane ki ijazat band	`powershell -ExecutionPolicy Bypass -File .\database\load.ps1`
Floor plan save par laal message	Koi cheez naqshe se bahar	Message mein naam likha hota hai, us ko andar karein
Raasta: "Start point is not inside..."	Click kamre/corridor se bahar	Room/Corridor ke andar click karein

---

19. Kya Kabhi Na Karein

❌ Na karein	Kyun
Ise asli fire alarm ki jagah lagana	Ye test aur lab mein check hua hai, certified nahi
Bina Safe Test Mode ke aag ka test	Asli logon ko asli SMS/Email chale jayenge
Asli sprinkler/alarm hardware bina ijazat jorna	Paani ya ghabrahat se nuqsan
`.env` kisi ko bhejna ya internet par daalna	Is mein passwords aur keys hain
AI ke "kuch nahi mila" ko "sab mehfooz" samajhna	AI se galti ho sakti hai

❌ Na karein	Kyun
Camera ka password `config.json` mein likhna	Environment variable mein rakhein
Model files (`.onnx`) badalna	Fingerprint match nahi hoga, model band ho jayega

Yaad rakhein: aag mein bijli jaye to camera band ho sakte hain. Cameras ko UPS/PoE backup par rakhein, aur hamesha smoke detectors aur fire panel istemaal karein.

---

20. Test Checklist (Print Karke Tick Karein)

#	Test	Chapter	✅ / ❌	Note
1	Software install, versions dikhe	2		
2	`.env` , database, packages, AI setup	3		
3	Safe Test Mode on	4		
4	Dono windows chalin (Server 5291, AI service 8008)	5		
5	Login hua, "Live" hara	6		
6	Simulator: Ground Floor - Warehouse mein aag, CRITICAL, incident, alarm lagatar, Ground Floor - Warehouse sprinklers khud chale	8		
6b	Simulator band: ~10 sec baad "Camera lost" (Ground Floor - Warehouse High)	8		
7	AI page: aag ki photo par laal dabba	9		
8	AI page: bina aag photo par aag nahi	9		
9	Sensor 50 = Warning, 60 = High, 75 = Critical	10		
10	Sprinkler: Manual wala khud nahi chala	11		
11	Emergency Dispatch + Resolve	12		

#	Test	Chapter	✅ / ❌	Note
12	Naqsha bana aur save hua	13		
13	Raasta + waqt dikha	13		
14	Aag mein raasta khud badla	13		
14b	Aag par risk heatmap aur "Next at risk", deewar ke peeche wala kamra mehfooz	13		
14c	Camera calibrate hua, naqshe par amber nuqte, nuqte se raasta	13		
15	History mein graph aur logs	14		
16	AI video camera: SIMULATED incident	15		
17	Browser notification aaya	16		
18	Automatic tests: Django 79 + AI service 25, aur 10 browser tests pass	17		
19	Safe Test Mode **band** kiya (`#` hataye)	4		
20	`save.ps1` → push → doosre PC par pull → `load.ps1`	5		
21	`start.ps1` se dono windows khulin	5		
22	Devices page: camera add, edit, delete (Admin)	15		

Tester ka naam: ..... **Tareekh:** .....